COSC3106: Theory of Computing

Lec#6: NFA Closure Properties

School of Computer Science and Technology

This lecture

- By the end of this lecture, you will learn …
  - NFA closure under the regular operations

Closure under the regular operations

- Theorem 2.1:
  - A language A is regular if and only if there exists a nondeterministic finite automaton NFA that accepts A.

- We have explained why DFA makes it not clear that the concatenation of two regular languages is regular, and that the star of a regular language is regular.

- Now, we will see that the concept of NFA can be used to give a simple proof of the fact that the regular languages are indeed closed under the regular operations.

Closure under the regular operations

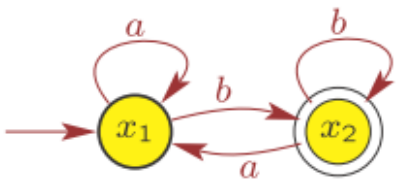
- Now, we will see that the concept of NFA can be used to give a simple proof of the fact that the regular languages are indeed closed under the regular operations.

- Theorem 2.2:
  - The set of regular languages is closed under the union operation
  - i.e., if A_1 and A_2 are regular languages over the same alphabet Σ, then $A_1 \cup A_2$ is also a regular language.

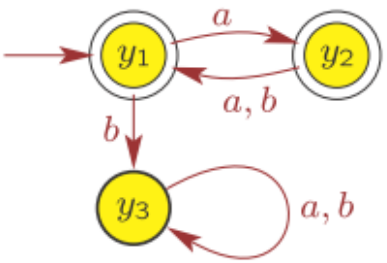
- Proof:

- Let us recall this just to see what we have done for DFA:
  - DFA M_1 accepts language $A_1 = L(M_1)$
  - DFA M_2 accepts language $A_2 = L(M_2)$

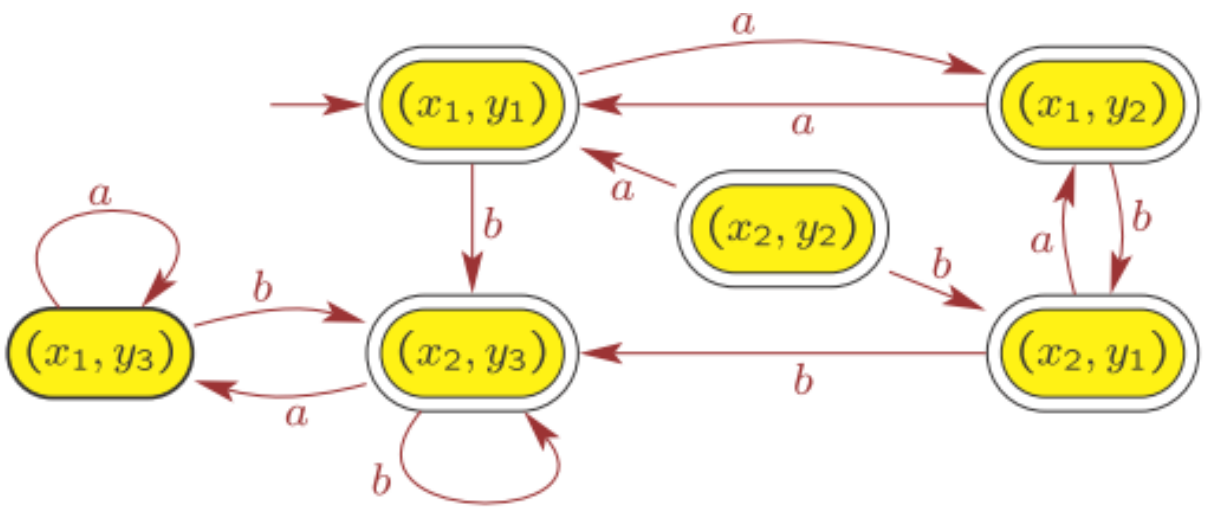
DFA M_1 for A_1



DFA M_2 for A_2



- We now want create a DFA M_3 for $A_1 \cup A_2$.



Closure under the regular operations

- Now, we will see that the concept of NFA can be used to give a simple proof of the fact that the regular languages are indeed closed under the regular operations.

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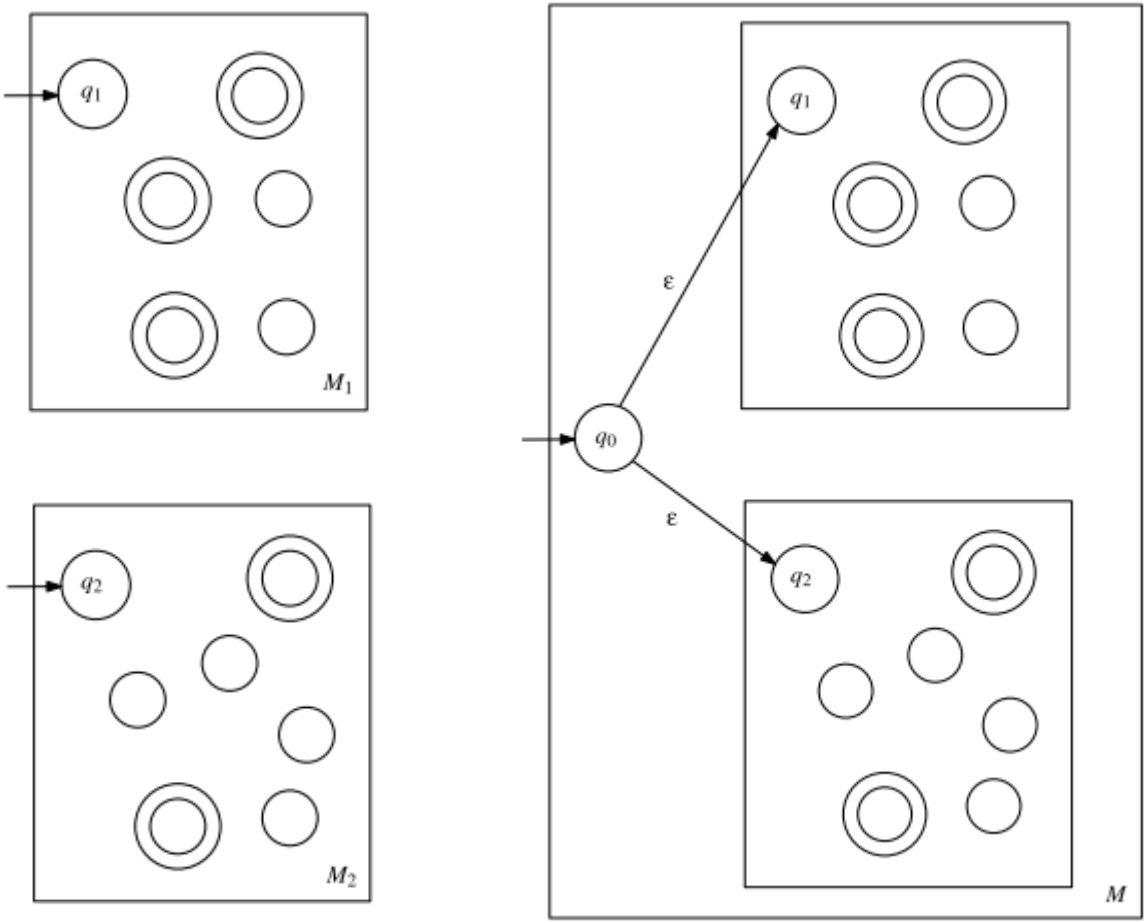
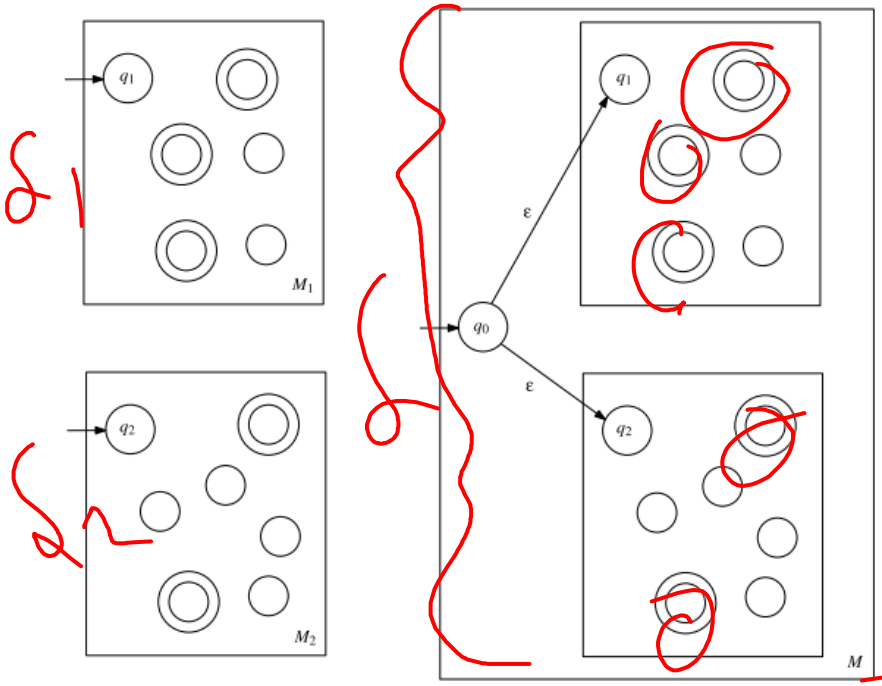


Closure under the regular operations



$q = \{q_0\}$

$Q = Q_1 \cup Q_2$

$\delta : Q \times \Sigma_\varepsilon \to P(Q)$

$\delta(r, a) \to ?$

If $r \in Q_1 : \delta(r,a) = \delta_1(r,a)$

$r \in Q_2 : \delta(r,a) = \delta_2(r,a)$

$r = q_0 \}$ $a = \varepsilon$ $\delta(r,a) = \{q_1, q_2\}$

$a \neq \varepsilon$ $\delta(r,a) = \{\emptyset\}$

$F = F_1 \cup F_2$

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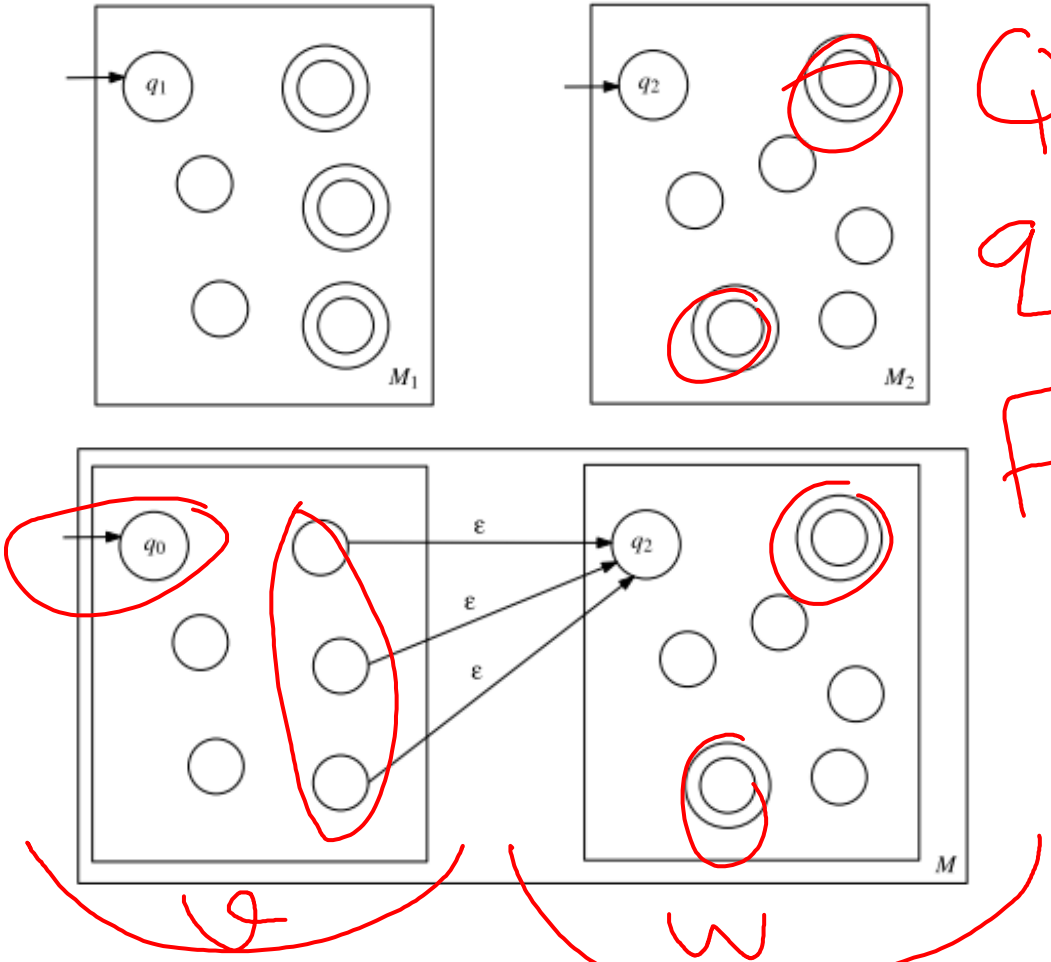
- **Theorem 2.3:**
  - The set of regular languages is closed under the concatenation operation,
  - i.e., if A_1 and A_2 are regular languages over the same alphabet Σ, then A_1A_2 is also a regular language.
    - Ex: Concatenation of A and B: $AB = \{vw \mid v \in A, w \in B\}$

- **Proof:**



$Q = Q_1 \cup Q_2$

$q = q_0$

$F = F_2$

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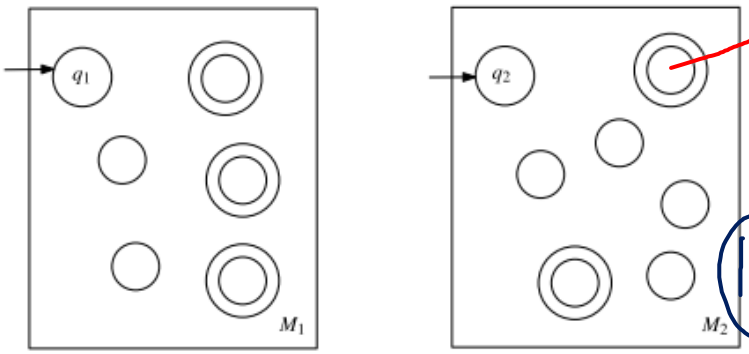
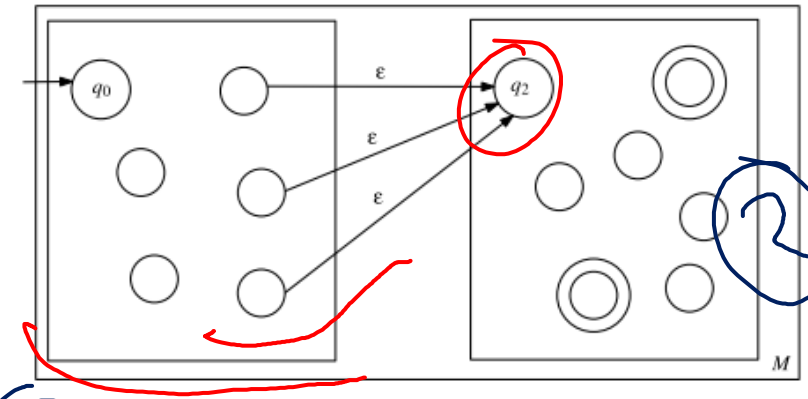
1 If $r \in Q_2 \Rightarrow \delta_2(r,a)$



$\delta(r,a) =$

1 if $r \in Q_1$ and $r \notin F_1$

$\Rightarrow \delta_1(r,a)$



2 if $r \in F_1$ and $a \neq \varepsilon$

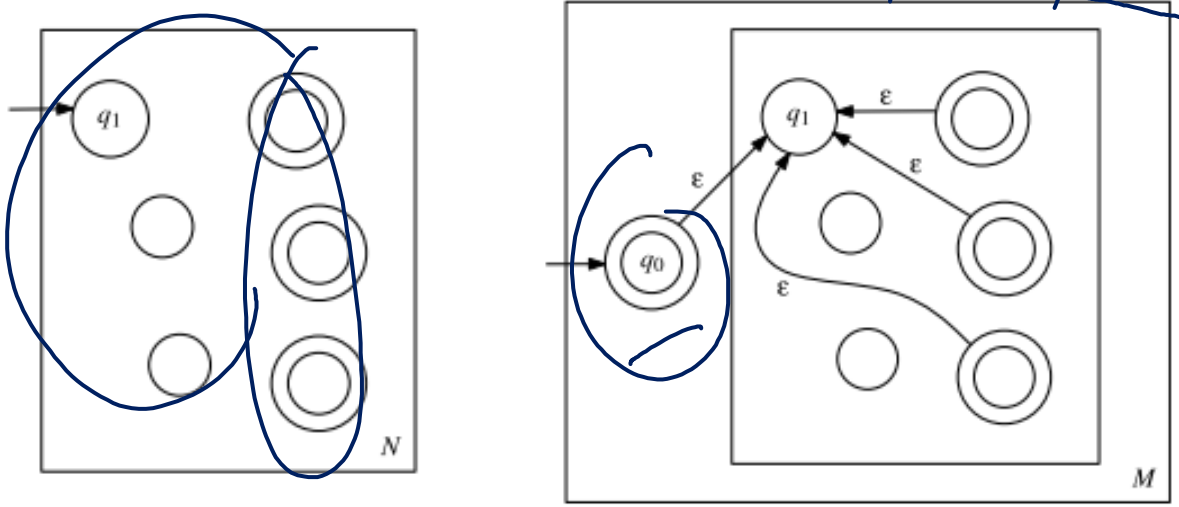
$\Rightarrow \delta_1(r,a)$

3 $r \in F_1$; $a = \varepsilon \Rightarrow \delta_1(r,a) \cup q_2$

- **Theorem 2.4:**
  - The set of regular languages is closed under the star operation,
  - i.e., if A is a regular language, then A^* is also a regular language.

- **Proof:**

$\boxed{u_1 | u_2 | u_3 | \cdots | \cdots | u_k}$

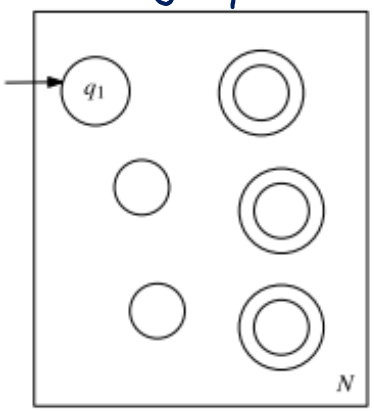


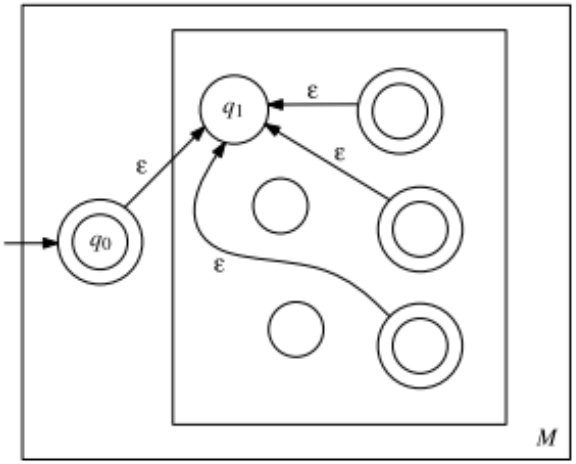
$Q = Q_1 \cup q_0$

$q = q_0$

$F = q_0 \cup F_1$

δ_1





1 if $r \in Q_1$ and $r \notin F_1 \Rightarrow \delta_1(r,a)$

2 $r \in F_1$ and $a \neq \varepsilon \Rightarrow \delta_1(r,a)$

3 $r \in F$, and $a = \varepsilon \Rightarrow \delta_1(r,a) \cup q_1$

up $r = q_0$ and $a = \varepsilon \Rightarrow q_1$

$a \neq \varepsilon \Rightarrow \emptyset$